

Air University



SSDD (Lab)

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Task # 01**Exercise 1****Answer the questions****1. A software bug is a?**

An error, flaw, or fault in a computer program that causes it to produce an incorrect or unexpected result.

2. The process of finding and fixing bugs is termed exception handling? t/f

False. (It is termed as **debugging**. Exception handling manages errors during execution.)

3. In software testing, the bug can occur for the?

Requirement, design, or code (essentially any stage of SDLC).

4. An impersonation of an authorized user is an example of a threat? t/f

True. (specifically known as spoofing or masquerading).

5. The records of each patient who is receiving or has received treatment resemble which security concept?

A. Assets

B. Threats

C. Vulnerability

D. Controls

Answer: Asset (information/data that has value to the organization).

6. Which of the following is not a protection system?

A. A system to stop a train if it passes a red light.

B. System to indicate non-return of library book.

C. System to shut down a reactor if the temperature/pressure are too high.

D. None of the mentioned.

Answer: A system to indicate non-return of a library book (this is a monitoring/tracking system, not a safety-critical protection system like others).

7. To specify the security requirements, one should identify the risks that are to be dealt with? t/f

True (Risk assessment is the foundation for security requirements).

8. Consider a system where heat sensors detect an intrusion and alert the security company. What kind of a requirement system is providing?

- A. Functional
- B. Non-Functional
- C. Known Requirement
- D. None of the mentioned

Answer: Functional (it describes a specific behavior or function the system must perform).

9. Choose the incorrect statement with respect to Non-Functional Requirement (NFR).

- A. Product-oriented Approach: Focus on system (or software) quality
- B. Process-oriented Approach: Focus on how NFRs can be used in the design process
- C. Quantitative Approach: Find measurable scales for the functionality attributes
- D. Qualitative Approach: Study various relationships between quality goals

Answer: Quantitative approach finds measurable scales for the functionality attributes.

10. Mention some NFRs of any system?

- Usability
- Reliability
- Performance
- Security
- Scalability

Exercise 2

Identify from the following statements whether its bug or an error?

- **If the end user clicks the “Save” button, but their entered data is not saved?**
Bug (the software failed to perform the intended function for the user).
- **A login button does not allow users to log in, an Add to cart button that does not update the cart, a “search box” not respond to a user query, etc.**
Bug (there are functional failures within the application).
- **A developer may misunderstand a design notation, or a programmer might type a variable name incorrectly, which may lead to -----**
Error (this refers to human action or mistakes that produce incorrect results).

- **Entering a significantly larger or a smaller number, or entering an input value of an undefined data type. These often pop up in form validations during functional testing of web or mobile apps.**

Error (these are often input or validation errors handled during testing).

- **If you sell a \$4 coffee to a customer and they go to pay for it, and the price displays as "- \$4", there's a flaw in the software that gives an unexpected result.**
Bug (a flaw in code logic resulting in incorrect behavior).

THE END
